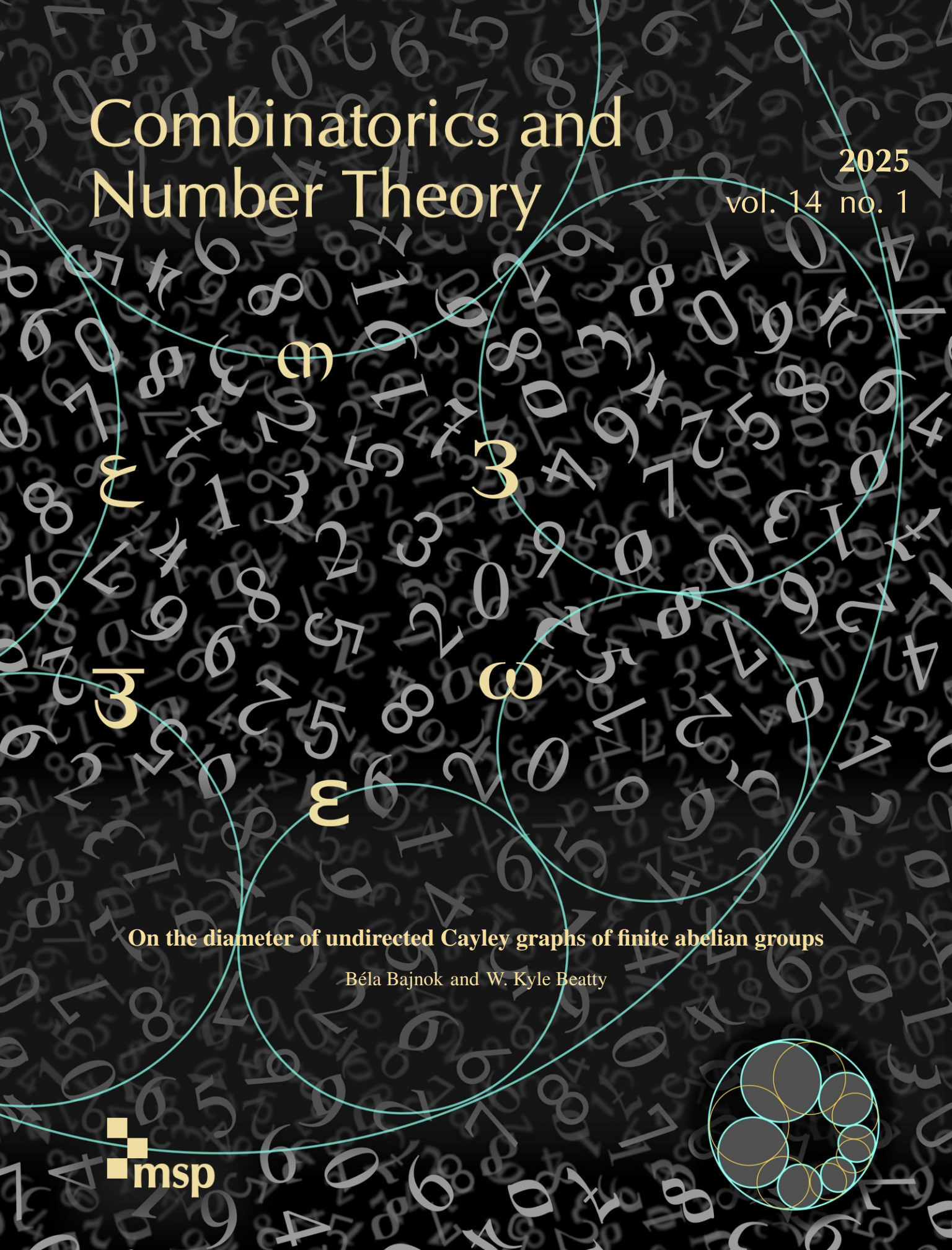
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# On the diameter of undirected Cayley graphs of finite abelian groups

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Let $s$ be a positive integer. Our goal is to find all finite abelian groups $G$ that contain a 2-subset $A$ for which the undirected Cayley graph $\Gamma(G, A)$ has diameter at most $s$. We provide a complete answer when $G$ is cyclic, and a conjecture and some partial answers when $G$ is noncyclic.

## 1. Introduction

Throughout this paper we let $G$ be an additively written finite abelian group. When $G$ is cyclic and of order $n$, we identify it with $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$; we consider $0, 1, \ldots, n-1$ interchangeably as integers and as elements of $\mathbb{Z}_n$. Below we use standard terminology and notation of additive combinatorics (see [Bajnok 2018]).

Given a finite abelian group $G$ and a subset $A \subseteq G$, the *undirected Cayley graph* $\Gamma(G, A)$ has its vertex set formed by the elements of $G$, and has an edge between two vertices whenever their difference is an element of $A$; that is, $g_1 \in G$ and $g_2 \in G$ are connected when $g_2 = g_1 + a$ or $g_1 = g_2 + a$ for some $a \in A$. If $A$ is a generating set of $G$, then $\Gamma(G, A)$ is connected; in particular, if $A$ is an *$s$-spanning set* of $G$, meaning that every element of $G$ can be written as the signed sum of at most $s$ (not-necessarily distinct) elements of $A$, then $\Gamma(G, A)$ has diameter at most $s$.

Suppose now that $A$ is an $m$-subset of $G$. It is easy to compute that there are at most $\sum_{i\geq 0}\binom{m}{i}\binom{h-1}{i-1}2^i$ vertices of $\Gamma(G, A)$ that are of distance $h$ away from any given vertex, and thus if $A$ is an $s$-spanning set in $G$, then the order of $G$ can be at most

$$\sum_{h=0}^{s}\sum_{i\geq 0}\binom{m}{i}\binom{h-1}{i-1}2^i = \sum_{i\geq 0}\binom{m}{i}\binom{s}{i}2^i.$$

Therefore, we have the following bound.

**Proposition 1.1.** *For positive integers $m$ and $s$, define*

$$a(m, s) = \sum_{i\geq 0}\binom{m}{i}\binom{s}{i}2^i. \tag{1}$$

*If an abelian group $G$ has an $m$-subset $A$ for which the diameter of $\Gamma(G, A)$ is $s$, then the order of $G$ is at most $a(m, s)$.*

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We note, in passing, that the values of $a(m, s)$ are known as *Delannoy numbers* [1895] and that they satisfy the recursion

$$a(m, s) = a(m, s-1) + a(m-1, s) + a(m-1, s-1).$$

It is a natural question to ask for the largest possible order of a group $G$ with an $m$-subset $A$ for which the diameter of $\Gamma(G, A)$ is at most $s$. Before we discuss our results on this question, let us review the analogous question in the directed case.

Given a finite abelian group $G$ and a subset $A \subseteq G$, the *directed Cayley graph* $\vec{\Gamma}(G, A)$ has its vertex set formed by the elements of $G$, and has a directed edge from $g_1 \in G$ to $g_2 \in G$ when $g_2 = g_1 + a$ for some $a \in A$. We say that $A$ is an *$s$-basis* of $G$ if every element of $G$ can be written as the sum of at most $s$ (not-necessarily distinct) elements of $A$; in this case, the diameter of $\vec{\Gamma}(G, A)$ is at most $s$.

For an $m$-subset $A$ of $G$, we see that there are at most $\binom{m+h-1}{h}$ vertices of $\vec{\Gamma}(G, A)$ that are of distance $h$ away from any given vertex, so if $A$ is an $s$-basis of $G$, then we must have

$$|G| \le \sum_{h=0}^{s} \binom{m+h-1}{h} = \binom{m+s}{s}.$$

When equality holds, we say that $A$ is a *perfect $s$-basis* of $G$: in this case, every element of $G$ arises as the sum of at most $s$ (not-necessarily distinct) elements of $A$ in a unique way. Trivially, the set of nonzero elements is a perfect 1-basis in any group $G$, and the 1-element set consisting of a generator of $G$ is a perfect $s$-basis in the cyclic group of order $s+1$. It turns out that there are no others:

**Theorem 1.2** (Bajnok, Berson, and Just [Bajnok et al. 2022]). *If a subset $A$ of a finite abelian group $G$ is a perfect $s$-basis, then $s = 1$ and $A = G \setminus \{0\}$, or $G \cong \mathbb{Z}_{s+1}$ and $|A| = 1$.*

Given the scarcity of perfect bases, it is natural to ask for the largest groups with a given diameter and size of generating set. The paper [Mask et al. 2011] asks the following questions:

- What is the largest possible order of an abelian group $G$ with an $m$-subset $A$ for which the diameter of $\vec{\Gamma}(G, A)$ is at most $s$?
- What is the largest possible order of a cyclic group $G$ with an $m$-subset $A$ for which the diameter of $\vec{\Gamma}(G, A)$ is at most $s$?

After a series of papers [Morillo et al. 1985; Fiol et al. 1987; Hsu and Jia 1994; Mask et al. 2011; Fiol 2013], these two questions have been completely solved for the case $m = 2$. We have the following two results.

**Theorem 1.3** (Morillo, Fiol, and Fàbrega [Morillo et al. 1985]). *Let $s$ be a positive integer. The largest possible order of a cyclic group $G$ with a 2-subset for which the diameter of $\vec{\Gamma}(G, A)$ is at most $s$ is given by $\lfloor (s^2 + 4s + 3)/3 \rfloor$.*

**Theorem 1.4** [Fiol 2013]. *Let $s$ be any positive integer. The largest possible order of an abelian group $G$ with a 2-subset for which the diameter of $\vec{\Gamma}(G, A)$ is at most $s$ is given by $\lfloor (s^2 + 4s + 4)/3 \rfloor$.*

Observe that the quantities in Theorems 1.3 and 1.4 agree, unless $s \equiv 1 \bmod 3$; for that case one can verify that the group $G = \mathbb{Z}_k \times \mathbb{Z}_{3k}$ has the $s$-basis $\{(0, 1), (1, 3k-1)\}$, where $k = (s+2)/3$. However, the question specifically for noncyclic groups is still open when $s \not\equiv 1 \bmod 3$.

Let us now return to the undirected case. When equality occurs in Proposition 1.1, we call $A$ a *perfect $s$-spanning set* in $G$. Perfect spanning sets can be easily characterized for $m = 1$ and $s = 1$. Note that $a(1, s) = 2s + 1$ and, clearly, for any $a \in \mathbb{Z}_{2s+1}$ with $\gcd(a, 2s + 1) = 1$, the set $\{a\}$ is a perfect $s$-spanning set in $\mathbb{Z}_{2s+1}$. Also, $a(m, 1) = 2m + 1$, and for any group $G$ of order $2m + 1$, every asymmetric subset of size $m$ (that is, a set containing exactly one of any nonzero element or its inverse) is a perfect 1-spanning set in $G$.

From (1) we get $a(2, s) = 2s^2 + 2s + 1$, and below we establish the following result.

**Theorem 1.5.** *Let $s$ be a positive integer. The largest possible order of a cyclic group $G$ with a 2-subset for which the diameter of $\Gamma(G, A)$ is at most $s$ is given by $2s^2 + 2s + 1$.*

For noncyclic groups the situation seems a bit more nuanced. We have the following conjecture.

**Conjecture 1.6.** *Let $s \geq 3$ be an integer.*

- *If $2s + 1$ is prime, then the largest noncyclic group with an $s$-spanning set of size 2 has order $2s^2$.*
- *If $2s + 1$ is composite and $p$ is its smallest prime divisor, then the largest noncyclic group with an $s$-spanning set of size 2 has order $2s^2 + 2s - (p^2 - 1)/2$.*

*In particular, for any noncyclic group $G$ with an $s$-spanning set of size 2, we have $|G| \leq 2s^2 + 2s - 4$.*

(We add that the largest noncyclic group with a 2-spanning set of size 2 is $\mathbb{Z}_3 \times \mathbb{Z}_3$.)

Below we prove that the quantities in Conjecture 1.6 are lower bounds.

**Theorem 1.7.** *Let $s$ be a positive integer.*

- *The largest noncyclic group with an $s$-spanning set of size 2 has order at least $2s^2$.*
- *If $2s + 1$ is composite and $p$ is any prime divisor of $2s + 1$, then the largest noncyclic group with an $s$-spanning set of size 2 has order at least $2s^2 + 2s - (p^2 - 1)/2$.*

Establishing upper bounds appears to be more difficult, and we only have the partial results that we summarize here. We first note that a noncyclic group that has a spanning set of size 2 must have rank 2 and thus is isomorphic to $\mathbb{Z}_c \times \mathbb{Z}_{kc}$ for some integers $c \geq 2$ and $k \geq 1$.

We say that the group $\mathbb{Z}_c \times \mathbb{Z}_{kc}$ is *$s$-regular* for some positive integer $s$, if it possesses an $s$-spanning set of the form $\{(1, u), (1, v)\}$ for some $u, v \in \mathbb{Z}_{kc}$. While we have no precise count on the number of $s$-regular groups, it appears that most groups of rank 2 that have an $s$-spanning set of size 2 are $s$-regular. For example, for $s = 10$, of the 105 groups of rank 2 that have an $s$-spanning set of size 2, 103 are $s$-regular. (The two exceptions are $\mathbb{Z}_8 \times \mathbb{Z}_{16}$ and $\mathbb{Z}_{10} \times \mathbb{Z}_{20}$.) In Theorem 3.1 below we prove, for every odd prime $c$ and positive integer $k$, that if $\mathbb{Z}_c \times \mathbb{Z}_{kc}$ has an $s$-spanning set at all, then it is $s$-regular.

Our results for upper bounds can be summarized as follows.

**Theorem 1.8.** *Let $s$ be a positive integer, and let $G = \mathbb{Z}_c \times \mathbb{Z}_{kc}$ be an $s$-regular group for some integers $c \geq 2$ and $k \geq 1$.*

- *If $c = 3$ and $s + 1$ is divisible by 3, then $|G| \leq 2s^2 + 1$.*
- *If $c$ is odd and $s$ is divisible by $c$, then $|G| \leq 2s^2 + s$, and if $2s + 1$ is divisible by $c$, then $|G| \leq 2s^2 + 2s - (c^2 - 1)/2$.*
- *In all other cases, $|G| \leq 2s^2$.*

As a consequence of our results, we can prove that, for $s \geq 3$, every $s$-regular abelian group of rank 2 has order at most $2s^2 + 2s - 4$, with equality if, and only if, it is isomorphic to $\mathbb{Z}_3 \times \mathbb{Z}_{(2s^2+2s-4)/3}$ for some $s \equiv 1 \bmod 3$ (see Theorem 3.3 below).

For $c = 2$ we have exact results, even without the assumption that the group is $s$-regular:

**Theorem 1.9.** *Let $k$ and $s$ be positive integers. The largest possible order of a group of the form $\mathbb{Z}_2 \times \mathbb{Z}_{2k}$ that has an $s$-spanning set of size 2 is $2s^2$ when $s$ is even, and $2s^2 - 2$ when $s$ is odd.*

In Section 4 below we provide some further results on groups of the form $\mathbb{Z}_2 \times \mathbb{Z}_{2k}$.

For $m \geq 3$, the largest order of abelian groups with an $m$-subset for which the diameter of their Cayley graphs is at most $s$ is generally unknown, though some small cases for cyclic groups were displayed in [Graham and Sloane 1980; Haanpää 2004; Hsu and Jia 1994].

Of particular interest are perfect spanning sets. An $m$-subset $A$ of $G$ is called a *perfect $s$-spanning set* if every element of $G$ arises as a signed sum of at most $s$ terms in a unique way; in this case we have $|G| = a(m, s)$. Above we mentioned perfect spanning sets for $m = 1$, for $s = 1$, and for $m = 2$; we believe that there are no others.

**Conjecture 1.10.** *If an $m$-subset $A$ of a finite abelian group $G$ is a perfect $s$-spanning set, then one of the following holds*:

- $m = 1$, $G \cong \mathbb{Z}_{2s+1}$, *and* $A = \{a\}$ *for some* $a \in \mathbb{Z}_{2s+1}$ *with* $\gcd(a, 2s + 1) = 1$;
- $s = 1$ *and $G$ is the disjoint union of* $\{0\}$, $A$, *and* $-A$; *or*
- $m = 2$, $G \cong \mathbb{Z}_{2s^2+2s+1}$, *and* $A = d \cdot \{s, s + 1\}$ *for some* $d \in \mathbb{Z}_{2s+1}$ *with* $\gcd(d, 2s^2 + 2s + 1) = 1$.

In Sections 2 and 3 below, we present our results on lower bounds and upper bounds, respectively. In Section 4 we treat the case of the groups $\mathbb{Z}_2 \times \mathbb{Z}_{2k}$ in more detail.

## 2. Constructions and lower bounds

In this section we provide some explicit constructions for $s$-spanning sets of size 2. We denote the $s$-span of a 2-subset $A = \{a_1, a_2\}$ of $G$ by

$$\langle A \rangle_s = \{\lambda_1 a_1 + \lambda_2 a_2 \mid \lambda_1, \lambda_2 \in \mathbb{Z},\ |\lambda_1| + |\lambda_2| \leq s\}.$$

It will be helpful to have the following facts available.

**Lemma 2.1.** *For a given positive integer $s$, let*

$$I(s) = \{(\lambda_1, \lambda_2) \mid \lambda_1, \lambda_2 \in \mathbb{Z},\ |\lambda_1| + |\lambda_2| \leq s\}.$$

(1) *For a given integer $h$ with $|h| \leq s$, the number of ordered pairs $(\lambda_1, \lambda_2) \in I(s)$ with $\lambda_1 + \lambda_2 = h$ equals $s + 1$ when $s$ and $h$ have the same parity, and $s$ otherwise.*

(2) *There are $(s + 1)^2$ ordered pairs $(\lambda_1, \lambda_2) \in I(s)$ where $\lambda_1 + \lambda_2$ and $s$ have the same parity, and $s^2$ where $\lambda_1 + \lambda_2$ and $s$ have opposite parity.*

(3) *There are $2s^2 + 2s + 1$ elements in $I(s)$.*

*Proof.* By symmetry, we may assume that $h \geq 0$. Observe that the "leftmost" integer point $(\lambda_1, \lambda_2)$ on the line $x + y = h$ within $|\lambda_1| + |\lambda_2| \leq s$ is

$$\left(-\left\lfloor \tfrac{1}{2}(s-h) \right\rfloor, h + \left\lfloor \tfrac{1}{2}(s-h) \right\rfloor\right),$$

and the "rightmost" such point is

$$\left(\left\lfloor \tfrac{1}{2}(s+h) \right\rfloor, h - \left\lfloor \tfrac{1}{2}(s+h) \right\rfloor\right),$$

yielding a set of $s+1$ points when $s$ and $h$ have the same parity, and $s$ points otherwise. This establishes the first claim, from which our other two claims follow easily. □

**Proposition 2.2.** *For all positive integers $s$ and $n$ with*

$$2 \leq n \leq 2s^2 + 2s + 1,$$

*the set $\{s, s+1\}$ is an $s$-spanning set in $\mathbb{Z}_n$.*

*Proof.* The fact that $n$ cannot be larger than $2s^2 + 2s + 1$ follows from Proposition 1.1. We verify that, for every $n$ in the range given above, we have $\langle \{s, s+1\} \rangle_s = \mathbb{Z}_n$.

Considering the elements of

$$S = \{\lambda_1 s + \lambda_2(s+1) \mid \lambda_1, \lambda_2 \in \mathbb{Z},\ |\lambda_1| + |\lambda_2| \leq s\}$$

in $\mathbb{Z}$ (rather than $\mathbb{Z}_n$), we see that they lie in the interval $[-(s^2+s), (s^2+s)]$. Since the index set

$$I(s) = \{(\lambda_1, \lambda_2) \mid \lambda_1, \lambda_2 \in \mathbb{Z},\ |\lambda_1| + |\lambda_2| \leq s\}$$

contains exactly $2s^2 + 2s + 1$ elements (see Lemma 2.1), it suffices to prove that no integer in the interval $[-(s^2+s), (s^2+s)]$ can be written as an element of $S$ in two different ways.

For that, suppose that

$$\lambda_1 s + \lambda_2(s+1) = \lambda_1' s + \lambda_2'(s+1)$$

for some $(\lambda_1, \lambda_2) \in I(s)$ and $(\lambda_1', \lambda_2') \in I(s)$; without loss of generality, we can assume that $\lambda_2 \geq \lambda_2'$. Our equation implies that $\lambda_2 - \lambda_2'$ is divisible by $s$ and is, therefore, equal to $0,\ s$, or $2s$.

If $\lambda_2 - \lambda_2' = 2s$, then $\lambda_2 = s$ and $\lambda_2' = -s$; since $|\lambda_1| + |\lambda_2| \leq s$ and $|\lambda_1'| + |\lambda_2'| \leq s$, this can only happen if $\lambda_1 = \lambda_1' = 0$, but this case leads to a contradiction with our equation.

Assume, next, that $\lambda_2 - \lambda_2' = s$. Our equation then yields $\lambda_1 - \lambda_1' = -s - 1$. In this case, we have $\lambda_1 \leq 0,\ \lambda_1' \geq 0,\ \lambda_2 \geq 0$, and $\lambda_2' \leq 0$, and we see that

$$|\lambda_1| + |\lambda_2| = -\lambda_1 + \lambda_2 = (s + 1 - \lambda_1') + (s + \lambda_2') = 2s + 1 - (|\lambda_1'| + |\lambda_2'|) \geq s + 1,$$

but that is a contradiction.

This leaves us with the case $\lambda_2 = \lambda_2'$, which implies $\lambda_1 = \lambda_1'$, as claimed, and therefore the set $\{s, s+1\}$ is $s$-spanning in $\mathbb{Z}_n$. □

**Proposition 2.3.** *For every positive integer $s$, the set $\{(0, 1), (1, 1)\}$ is an $s$-spanning set in $\mathbb{Z}_s \times \mathbb{Z}_{2s}$.*

*Proof.* We need to prove that, for any $(a, b) \in \mathbb{Z}_s \times \mathbb{Z}_{2s}$, we have $(a, b) \in \langle \{(0, 1), (1, 1)\} \rangle_s$. Note that by symmetry we may assume that $0 \leq b \leq s$. We consider three subcases.

When $b \geq a$, we let $\lambda_1 = b - a$ and $\lambda_2 = a$. We then have

$$\lambda_1(0,1) + \lambda_2(1,1) = (a,b) \quad \text{and} \quad |\lambda_1| + |\lambda_2| = \lambda_1 + \lambda_2 = b \leq s,$$

so $(a,b) \in \langle\{(0,1),(1,1)\}\rangle_s$.

Next, when $2a - s \leq b < a$, we again let $\lambda_1 = b - a$ and $\lambda_2 = a$. In this case, we have

$$|\lambda_1| + |\lambda_2| = -\lambda_1 + \lambda_2 = 2a - b \leq s,$$

so again $(a,b) \in \langle\{(0,1),(1,1)\}\rangle_s$.

Finally, when $b < 2a - s$, we let $\lambda_1 = b - a + s$ and $\lambda_2 = a - s$. Here we have

$$\lambda_1(0,1) + \lambda_2(1,1) = (a - s, b),$$

where $a - s \equiv a \mod s$. Furthermore,

$$|\lambda_1| + |\lambda_2| = \lambda_1 - \lambda_2 = (b - a + s) + (s - a) < s,$$

so again $(a,b) \in \langle\{(0,1),(1,1)\}\rangle_s$. This completes our proof. □

**Theorem 2.4.** *Let $c$ and $s$ be positive integers such that the remainder of $s \mod c$ is at least $(c-1)/2$. Let $u = \lfloor s/c \rfloor$, $v = \lceil s/c \rceil$, and $k = 2uv$. Then $A = \{(1,u),(1,v)\}$ is an $s$-spanning set in $G = \mathbb{Z}_c \times \mathbb{Z}_{ck}$.*

*Proof.* Let $H = \langle(1,u)\rangle$ be the subgroup of $G$ generated by the element $(1,u)$. Since $H$ has order $2cv$, it has $cu$ cosets in $G$. We claim that these cosets are $\lambda_2 \cdot (1,v) + H$, where $\lambda_2 = 0, 1, \ldots, cu - 1$. In order to prove this, we need to show that these $cu$ cosets are pairwise distinct.

Suppose, then, that $\mu_1(1,v)$ and $\mu_2(1,v)$ are in the same coset of $H$ for some $0 \leq \mu_1, \mu_2 \leq cu - 1$. This implies that $(\mu_2 - \mu_1) \cdot (1,v)$ is an element of $H$, so there exists an integer $\mu$ for which

$$(\mu_2 - \mu_1) \cdot (1,v) = \mu \cdot (1,u).$$

From the first coordinates, this means that $\mu_2 - \mu_1 - \mu$ is divisible by $c$, and from the second coordinates we see that $(\mu_2 - \mu_1)v - \mu u$, which, since $v = u + 1$, equals $(\mu_2 - \mu_1 - \mu)u + (\mu_2 - \mu_1)$, is divisible by $ck$ and thus by $cu$. Combining these, we get that $\mu_2 - \mu_1$ must also be divisible by $cu$, but that is only possible if $\mu_1 = \mu_2$.

Let $g \in G$ be arbitrary. According to our claim, we have integers $\lambda_1 \in [-cv+1, cv]$ and $\lambda_2 \in [0, cu-1]$ for which

$$g = \lambda_1(1,u) + \lambda_2(1,v).$$

By assumption, we have integers $q$ (quotient) and $r$ (remainder) with $(c-1)/2 \leq r \leq c-1$ for which $s = qc + r$. We then have

$$|\lambda_1| + |\lambda_2| \leq cv + (cu - 1) = c(q+1) + (cq - 1) = 2cq + c - 1 = 2s + (c - 2r - 1) \leq 2s. \qquad (2)$$

If $|\lambda_1| + |\lambda_2| \leq s$, then $g \in \langle A\rangle_s$, and we are done.

Assume then that $|\lambda_1| + |\lambda_2| \geq s + 1$. In this case, let $\lambda_1' = \lambda_1 - \epsilon cv$ and $\lambda_2' = \lambda_2 - cu$, where $\epsilon$ equals 1 when $\lambda_1 \geq 0$, and $-1$ when $\lambda_1 < 0$. We can compute that

$$\lambda_1'(1,u) + \lambda_2'(1,v) = (\lambda_1 - \epsilon cv)(1,u) + (\lambda_2 - cu)(1,v) = \lambda_1(1,u) + \lambda_2(1,v) - (\epsilon cv + cu, (\epsilon+1)cuv),$$

and since $\epsilon + 1$ equals 0 or 2, in $G$ we have

$$\lambda_1'(1,u) + \lambda_2'(1,v) = \lambda_1(1,u) + \lambda_2(1,v) = g.$$

Finally, we show that $|\lambda_1'| + |\lambda_2'| \leq s$. Indeed, we have

$$|\lambda_1'| + |\lambda_2'| = |\lambda_1 - \epsilon cv| + |\lambda_2 - cu| = cv - |\lambda_1| + cu - |\lambda_2|.$$

From (2) above we see that $cv + cu \leq 2s + 1$, and by our current assumption we have $|\lambda_1| + |\lambda_2| \geq s + 1$. Therefore, $|\lambda_1'| + |\lambda_2'| \leq s$, as claimed, and our proof is now complete. □

**Corollary 2.5.** *Let $s$ be a positive integer for which $2s + 1$ is composite, and let $p$ be any prime divisor of $2s + 1$. Then the largest noncyclic group with an $s$-spanning set of size 2 has order at least $2s^2 + 2s - (p^2 - 1)/2$.*

*Proof.* Let us write $2s + 1 = pt$ for some integer $t \geq 2$. Note that $p$ and $t$ must be odd, and we have

$$s = \tfrac{1}{2}(t - 1)p + \tfrac{1}{2}(p - 1),$$

$\lfloor s/p \rfloor = (t - 1)/2$, and $\lceil s/p \rceil = (t + 1)/2$. Let $G = \mathbb{Z}_p \times \mathbb{Z}_{pk}$, where $k = (t^2 - 1)/2$. Then $G$ has order $2s^2 + 2s - (p^2 - 1)/2$ and, according to Theorem 2.4, it has an $s$-spanning set of size 2. □

## 3. $s$-regular groups and upper bounds

Recall that we say the group $\mathbb{Z}_c \times \mathbb{Z}_{kc}$ is *$s$-regular* for some positive integer $s$ if it possesses an $s$-spanning set of the form $\{(1, u), (1, v)\}$ for some $u, v \in \mathbb{Z}_{kc}$. In this section we provide some upper bounds for the largest possible order of an $s$-regular group.

**Theorem 3.1.** *Let $G = \mathbb{Z}_c \times \mathbb{Z}_{ck}$ for some integers $c \geq 2$ and $k \geq 1$, and let $G$ have an $s$-spanning set for some positive integer $s$. If $c$ is an odd prime, then $G$ is $s$-regular.*

*Proof.* First, we prove that $G$ has an $s$-spanning set of the form $A = \{(a, u), (b, v)\}$ for some $a, b \in \mathbb{Z}_c$ and $u, v \in \mathbb{Z}_{ck}$, with $u \not\equiv v \bmod c$. We note that if $\{(a, u), (b, v)\}$ is an $s$-spanning set, then so is $\{(a, u), (-b, -v)\}$, so if our claim were false, then we would have $u \equiv v \equiv -v \bmod c$ for each $s$-spanning set $A = \{(a, u), (b, v)\}$ of $G$. But since $c$ is an odd prime, that would imply that $u \equiv v \equiv 0 \bmod c$, and that is a contradiction with $A$ being a spanning set.

Suppose then that $A = \{(a, u), (b, v)\}$ is an $s$-spanning set in $G$ with $u \not\equiv v \bmod c$. We will prove that, in this case, $B = \{(1, u), (1, v)\}$ is also an $s$-spanning set in $G$.

Let

$$I(s) = \{(\lambda_1, \lambda_2) \mid \lambda_1, \lambda_2 \in \mathbb{Z},\ |\lambda_1| + |\lambda_2| \leq s\},$$

and consider the function $f : I(s) \to G$ given by

$$(\lambda_1, \lambda_2) \mapsto \lambda_1(a, u) + \lambda_2(b, v).$$

Since $A$ is an $s$-spanning set, $f$ is surjective; we can thus find a subset $I_0$ of $I(s)$ on which the restriction $f_0 : I_0 \to G$ of $f$ is bijective. We will prove that $g_0 : I_0 \to G$, given by

$$(\lambda_1, \lambda_2) \mapsto \lambda_1(1, u) + \lambda_2(1, v),$$

is also bijective. It suffices to show that $g_0$ is injective. In order to do so, let us assume that the equation

$$\lambda_1(1, u) + \lambda_2(1, v) = \mu_1(1, u) + \mu_2(1, v)$$

holds in $G$ for some $(\lambda_1, \lambda_2)$ and $(\mu_1, \mu_2)$ of $I_0$. We then have

$$\lambda_1 + \lambda_2 \equiv \mu_1 + \mu_2 \bmod c, \tag{3}$$

and

$$\lambda_1 u + \lambda_2 v \equiv \mu_1 u + \mu_2 v \bmod ck. \tag{4}$$

From (4) we find

$$(\lambda_1 + \lambda_2)u + \lambda_2(v-u) \equiv (\mu_1 + \mu_2)u + \mu_2(v-u) \bmod ck;$$

considering (3), this gives us

$$\lambda_2(v-u) \equiv \mu_2(v-u) \bmod c.$$

Therefore, since $c$ is prime and $u \not\equiv v \bmod c$, we get $\lambda_2 \equiv \mu_2 \bmod c$. From (3) then we get $\lambda_1 \equiv \mu_1 \bmod c$, and thus

$$\lambda_1 a + \lambda_2 b \equiv \mu_1 a + \mu_2 b \bmod c. \tag{5}$$

But $f_0$ is an injection, so (5) and (4) can hold together only when $(\lambda_1, \lambda_2) = (\mu_1, \mu_2)$. □

**Theorem 3.2.** *Let $G = \mathbb{Z}_c \times \mathbb{Z}_{ck}$ for some integers $c \geq 2$ and $k \geq 1$, and suppose that $G$ is $s$-regular for some positive integer $s$.*

(1) *If $c$ is even, then $|G| \leq 2s^2$, and equality may occur only when $s \equiv 0 \bmod c$.*

(2) *If $c$ is odd and*

- *$s \equiv 0 \bmod c$, then $|G| \leq 2s^2 + s$;*
- *$s \equiv (c-1)/2 \bmod c$, then $|G| \leq 2s^2 + 2s - (c^2-1)/2$;*
- *$c = 3$ and $s \equiv 2 \bmod 3$, then $|G| \leq 2s^2 + 1$.*

*In all other cases, $|G| < 2s^2$.*

*Proof.* Let $A = \{(1, u), (1, v)\}$ be an $s$-spanning set in $G$, in which case

$$\{(\lambda_1 + \lambda_2, \lambda_1 u + \lambda_2 v) \mid (\lambda_1, \lambda_2) \in \mathbb{Z}^2,\ |\lambda_1| + |\lambda_2| \leq s\} = G.$$

Recall from Lemma 2.1 that, for a given integer $h$ with $|h| \leq s$, the number $\Lambda(h)$ of ordered pairs $(\lambda_1, \lambda_2) \in \mathbb{Z}^2$ with $|\lambda_1| + |\lambda_2| \leq s$ and $\lambda_1 + \lambda_2 = h$ equals $s+1$ when $s$ and $h$ have the same parity, and $s$ otherwise. For each $i = 0, 1, \ldots, c-1$, we let $N(i)$ denote the sum of all $\Lambda(h)$ with $|h| \leq s$ and $h \equiv i \bmod c$. Since $A$ is an $s$-spanning set in $G$, we must have

$$\{(\lambda_1 + \lambda_2, \lambda_1 u + \lambda_2 v) \mid (\lambda_1, \lambda_2) \in \mathbb{Z}^2,\ |\lambda_1| + |\lambda_2| \leq s\} = \mathbb{Z}_c \times \mathbb{Z}_{ck},$$

and thus, for each $i = 0, 1, \ldots, c-1$,

$$\{(\lambda_1 + \lambda_2, \lambda_1 u + \lambda_2 v) \mid (\lambda_1, \lambda_2) \in \mathbb{Z}^2,\ |\lambda_1| + |\lambda_2| \leq s,\ \lambda_1 + \lambda_2 \equiv i \bmod c\} = \{i\} \times \mathbb{Z}_{ck}.$$

In particular,

$$|G| \leq cN(i) \tag{6}$$

must hold for each $i = 0, 1, \ldots, c-1$. We now inspect (6) in several cases.

When $c$ is even, we consider $i = r+1$, where $r$ is the nonnegative remainder of $s \bmod c$ (with $r+1$ understood to be 0 when $r = c-1$). To compute $N(r+1)$, we need to identify the values of $h$ for which $|h| \le s$ and $h \equiv r+1 \bmod c$, and these values are $h = s+1-jc$ for $j = 1, 2, \ldots, j_{\max}$, where

$$j_{\max} = \left\lfloor \frac{2s+1}{c} \right\rfloor = \frac{2s-2r}{c} + \left\lfloor \frac{2r+1}{c} \right\rfloor.$$

Note that $\lfloor (2r+1)/c \rfloor$ equals 0 when $0 \le r \le c/2-1$, and 1 when $c/2 \le r \le c-1$. Since $s$ and $h = s+1-jc$ have different parity, and thus $\Lambda(h) = s$ for each $j$, we have

$$cN(r+1) = \left(2s - 2r + c\left\lfloor \frac{2r+1}{c} \right\rfloor\right)s,$$

and we see that this quantity is at most $2s^2$, with equality when $r = 0$ or $r = c/2$. But $|G|$ must be divisible by $c^2$, so $|G| = 2s^2$ may only occur when $r = 0$.

When $c$ is odd and $r \le c-2$, we consider $i = r+2$ (where $r+2$ is understood to be 0 when $r = c-2$). In this case the values of $h$ with $|h| \le s$ and $h \equiv r+2 \bmod c$ are $h = s+2-tc$ for $t = 1, 2, \ldots, t_{\max}$, where

$$t_{\max} = \left\lfloor \frac{2s+2}{c} \right\rfloor = \frac{2s-2r}{c} + \left\lfloor \frac{2r+2}{c} \right\rfloor.$$

Note that $\lfloor (2r+2)/c \rfloor$ equals 0 when $0 \le r \le (c-3)/2$, and 1 when $(c-1)/2 \le r \le c-2$. Since $s$ and $h = s+2-tc$ have the same parity when $t$ is even and different parity when $t$ is odd, we have

$$cN(r+2) = \left(s - r + c\left\lfloor \frac{2r+2}{c} \right\rfloor\right)s + (s-r)(s+1).$$

We consider four subcases. For $r = 0$, this quantity equals $2s^2 + s$, and when $1 \le r \le (c-3)/2$, it equals $(s-r)s + (s-r)(s+1)$, which is less than $2s^2$. When $(c+1)/2 \le r \le c-2$, we have

$$cN(r+2) = (s-r+c)s + (s-r)(s+1) = (s-r)(2s+1) + cs \le \left(s - \tfrac{1}{2}(c+1)\right)(2s+1) + cs,$$

which again is less than $2s^2$. Our last subcase is when $r = (c-1)/2$, when we have

$$cN(r+2) = \left(s - \tfrac{1}{2}(c-1)\right)(2s+1) + cs = 2s^2 + 2s - \tfrac{1}{2}(c-1).$$

Observe that when $s \equiv (c-1)/2 \bmod c$, then $2s^2 + 2s - (c^2-1)/2 \equiv 0 \bmod c^2$, and since $|G|$ must be divisible by $c^2$, this is the largest it can then be.

Our final case is when $c$ is odd and $r = c-1$; we consider $i = 0$. In this case the values of $h$ with $|h| \le s$ and $h \equiv 0 \bmod c$ are $h = s+1-zc$ for $z = 1, 2, \ldots, z_{\max}$, where

$$z_{\max} = \left\lfloor \frac{2s+1}{c} \right\rfloor = \frac{2s+2-c}{c}.$$

Note that $z_{\max}$ is odd. Since $s$ and $h = s+1-zc$ have the same parity when $z$ is odd and different parity when $z$ is even, we have

$$N(0) = \tfrac{1}{2}(z_{\max} - 1) \cdot s + \tfrac{1}{2}(z_{\max} + 1) \cdot (s+1),$$

and thus

$$cN(0) = (s+1-c)s + (s+1)(s+1) = 2s^2 - (c-3)s + 1,$$

which is less than $2s^2$, unless $c = 3$. This completes our proof. □

**Theorem 3.3.** *For $s \geq 3$, every $s$-regular abelian group of rank 2 has order at most $2s^2 + 2s - 4$, with equality if, and only if, it is isomorphic to $\mathbb{Z}_3 \times \mathbb{Z}_{(2s^2+2s-4)/3}$ for some $s \equiv 1 \mod 3$.*

*Proof.* Suppose that $G = \mathbb{Z}_c \times \mathbb{Z}_{ck}$ is an $s$-regular group of order at least $2s^2 + 2s - 4$ for some integers $s \geq 3$, $c \geq 2$, and $k \geq 1$. Then, by Theorem 3.2, $c$ is odd, so $c \geq 3$. Recalling that $|G| = c^2k$, we have

- $s \equiv 0 \mod c$ and $2s^2 + 2s - 4 \leq c^2k \leq 2s^2 + s$, or
- $s \equiv (c-1)/2 \mod c$ and $2s^2 + 2s - 4 \leq c^2k \leq 2s^2 + 2s - (c^2-1)/2$.

Since the first possibility leads to no solutions, our claim is established. □

## 4. The groups $\mathbb{Z}_2 \times \mathbb{Z}_{2k}$

In this section we provide some results for $s$-spanning sets in groups of the form $\mathbb{Z}_2 \times \mathbb{Z}_{2k}$, including the exact value of the largest possible order of such groups. Our general conjecture is as follows.

**Conjecture 4.1.** *Let $k$ and $s$ be positive integers. The group $\mathbb{Z}_2 \times \mathbb{Z}_{2k}$ has an $s$-spanning set of size 2 if, and only if, one of the following holds*:

- *$s$ is odd and $k \leq (s^2-1)/2$;*
- *$s$ is even and $k \leq (s^2-s)/2$;*
- *$s \equiv 0 \mod 4$, $k$ is even, and $(s^2-s)/2 < k \leq s^2/2$; or*
- *$s \equiv 2 \mod 4$, $k \equiv 2 \mod 4$, and $(s^2-s)/2 < k \leq s^2/2$.*

**Theorem 4.2.** *Let $k$ and $s$ be positive integers. If the group $G = \mathbb{Z}_2 \times \mathbb{Z}_{2k}$ has an $s$-spanning set of size 2, then it has order at most $2s^2$.*

*Proof.* If $G$ is $s$-regular, that is, it has an $s$-spanning set of the form $\{(1, u), (1, v)\}$ for some $u, v \in \mathbb{Z}_{2k}$, then our claim follows from Theorem 3.2. Since no set of the form $\{(0, u), (0, v)\}$ can be a spanning set, we can suppose that $G$ has an $s$-spanning set of the form $A = \{(0, u), (1, v)\}$ for some $u, v \in \mathbb{Z}_{2k}$. Therefore, each element in $\langle A \rangle_s$ has the form $(\lambda_2, \lambda_1 u + \lambda_2 v)$ for some integers $\lambda_1$ and $\lambda_2$ with $|\lambda_1| + |\lambda_2| \leq s$.

Clearly, if $u$ and $v$ are both even, then $A$ can only generate elements of $G$ with an even second component, and thus it cannot be a spanning set.

Now if $u$ and $v$ are both odd, then $\lambda_1 u + \lambda_2 v$ has the same parity as $\lambda_1 + \lambda_2$. According to Lemma 2.1, the parity of $\lambda_1 + \lambda_2$ differs from the parity of $s$ exactly $s^2$ times (and agrees with it $(s+1)^2$ times) within $|\lambda_1| + |\lambda_2| \leq s$. This means that $A$ cannot generate more than $s^2$ elements in $\mathbb{Z}_2 \times \mathbb{Z}_{2k}$ whose second coordinate has different parity than $s$ does, so if it is an $s$-spanning set, then we must have $|G| \leq 2s^2$.

Next, if $u$ is even and $v$ is odd, then for each element of $G$ that $A$ generates, its two components will have the same parity (namely, the parity of $\lambda_2$). Therefore, $A$ cannot be a spanning set.

Finally, we prove that if $A = \{(0, u), (1, v)\}$ is an $s$-spanning set of $G$ with $u$ odd and $v$ even, then $A' = \{(1, u), (1, v)\}$ is an $s$-spanning set as well, meaning that $G$ is $s$-regular and thus has order at most $2s^2$ by Theorem 3.2.

Let $a \in \mathbb{Z}_2$ and $b \in \mathbb{Z}_{2k}$. Since $\langle A \rangle_s = G$, there are integers $\lambda_1$ and $\lambda_2$ with $|\lambda_1| + |\lambda_2| \leq s$ for which

$$\lambda_1(0, u) + \lambda_2(1, v) = (\lambda_2, \lambda_1 u + \lambda_2 v) = (a', b),$$

where $a' = a$ if $b$ is even and $a' = 1 + a$ when $b$ is odd. Observe that when $b$ is even, since $u$ is odd and $v$ is even, $\lambda_1$ must be even, and thus

$$\lambda_1(1, u) + \lambda_2(1, v) = (\lambda_1 + \lambda_2, \lambda_1 u + \lambda_2 v) = (\lambda_2, \lambda_1 u + \lambda_2 v) = (a', b) = (a, b).$$

Similarly, when $b$ is odd, $\lambda_1$ must be odd, and

$$\lambda_1(1, u) + \lambda_2(1, v) = (1 + a', b) = (a, b).$$

Therefore, $\langle A' \rangle_s = G$, as claimed, which completes our proof. □

**Theorem 4.3.** *Let $k$ and $s$ be positive integers with $s$ odd. Then the group $G = \mathbb{Z}_2 \times \mathbb{Z}_{2k}$ has an $s$-spanning set of size 2 if, and only if, $k \leq (s^2 - 1)/2$.*

*Proof.* The fact that $k \leq (s^2 - 1)/2$ is necessary follows from Theorem 4.2, as $s$ is odd. We may also recall that Theorem 2.4, applied with $c = 2$ and odd $s$, yields that, with $u = (s - 1)/2$ and $v = (s + 1)/2$, the set $\{(1, u), (1, v)\}$ is an $s$-spanning set in $\mathbb{Z}_2 \times \mathbb{Z}_{s^2-1}$. This takes care of the case $k = (s^2 - 1)/2$.

Our goal below is to show that the same values of $u = (s - 1)/2$ and $v = (s + 1)/2$ as *integers* provide an $s$-span for the set $\mathbb{Z}_2 \times \{0, 1, 2, \ldots, k\}$ when $k \leq (s^2 - 3)/2$. (Here we consider the operation in the first component in $\mathbb{Z}_2$ and in the second component in $\mathbb{Z}$.) If $u$ and $v$ span $\mathbb{Z}_2 \times \{1, 2, \ldots, k - 1\}$, then by symmetry they will also span $\mathbb{Z}_2 \times \{-k + 1, -k + 2, \ldots, -1\}$, which correspond to group elements $\mathbb{Z}_2 \times \{k + 1, k + 2, \ldots, 2k - 1\} \subset \mathbb{Z}_2 \times \mathbb{Z}_{2k}$. We can then conclude that $\mathbb{Z}_2 \times \mathbb{Z}_{2k}$ has an $s$-spanning set of size 2 for $k \leq (s^2 - 3)/2$ as well.

We thus need to establish that, for every $a \in \mathbb{Z}_2$ and $b \in \{0, 1, 2, \ldots, k\}$, there are integers $\mu_1$ and $\mu_2$ with $|\mu_1| + |\mu_2| \leq s$ for which

$$\mu_1\big(1, \tfrac{1}{2}(s - 1)\big) + \mu_2\big(1, \tfrac{1}{2}(s + 1)\big) = (a, b) \tag{7}$$

holds in $\mathbb{Z}_2 \times \{0, 1, 2, \ldots, k\}$.

Let us recall from the proof of Theorem 2.4 (applied again for odd $s$ and $c = 2$) that there are integers $\lambda_1 \in [-s, s + 1]$ and $\lambda_2 \in [0, s - 2]$ for which one of the following two sets of conditions holds:

(i) $|\lambda_1| + |\lambda_2| \leq s$ and (7) is satisfied in $\mathbb{Z}_2 \times \mathbb{Z}_{s^2-1}$ with $(\mu_1, \mu_2) = (\lambda_1, \lambda_2)$, or

(ii) $|\lambda_1| + |\lambda_2| \geq s + 1$ and (7) is satisfied in $\mathbb{Z}_2 \times \mathbb{Z}_{s^2-1}$ with $(\mu_1, \mu_2) = (\lambda_1 - \epsilon(s + 1), \lambda_2 - s + 1)$, where $\epsilon$ equals 1 when $\lambda_1 \geq 0$, and $-1$ when $\lambda_1 < 0$.

To complete our proof, we will show that, considering

$$B = \mu_1 \tfrac{1}{2}(s - 1) + \mu_2 \tfrac{1}{2}(s + 1)$$

as an integer (rather than as an element of $\mathbb{Z}_{s^2-1}$), we have $B = b$. Since $0 \leq b \leq k < (s^2 - 1)/2$, it suffices to prove that

$$-\tfrac{1}{2}(s^2 - 1) \leq B < s^2 - 1. \tag{8}$$

In case (i), we have $\mu_1 = \lambda_1 \in [-s, s + 1]$ and $\mu_2 = \lambda_2 \in [0, s - 2]$, from which (8) readily follows. For case (ii), we find that

$$|\mu_1| + |\mu_2| = (s + 1 - |\lambda_1|) + (s - 1 - |\lambda_2|) \leq 2s - (s + 1) = s - 1,$$

and thus

$$|B| = \left|\mu_1 \tfrac{1}{2}(s-1) + \mu_2 \tfrac{1}{2}(s+1)\right| \le |\mu_1|\tfrac{1}{2}(s-1) + |\mu_2|\tfrac{1}{2}(s+1) \le (|\mu_1| + |\mu_2|)\tfrac{1}{2}(s+1) \le \tfrac{1}{2}(s^2-1),$$

so (8) holds again. This completes our proof. □

**Theorem 4.4.** *Let $k$ and $s$ be positive integers with $s$ even. Then the group $G = \mathbb{Z}_2 \times \mathbb{Z}_{2k}$ has an $s$-spanning set of size 2 if $k \le (s^2 - s)/2$.*

*Proof.* Like in the proof of Theorem 4.3, we will prove that there are integers $u$ and $v$ for which the $s$-span of $A = \{(1, u), (1, v)\}$ contains the set $\mathbb{Z}_2 \times \{0, 1, 2, \ldots, k\}$. (Here too we consider the operation in the first component in $\mathbb{Z}_2$ and in the second component in $\mathbb{Z}$.)

Since $s - 1$ is odd, from the proof of Theorem 4.3, with $u = (s-2)/2$ and $v = s/2$, we see that $\langle A\rangle_{s-1}$ contains the set $\mathbb{Z}_2 \times \{0, 1, 2, \ldots, k\}$ when $k \le ((s-1)^2 - 1)/2$. Therefore, $(1, v) + \langle A\rangle_{s-1}$ contains $\mathbb{Z}_2 \times \{s/2, s/2+1, \ldots, s/2+k\}$ for each $k \le (s^2 - 2s)/2$, and since $(1, v) + \langle A\rangle_{s-1} \subseteq \langle A\rangle_s$, our claim follows for all $k \le (s^2 - s)/2$. □

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